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COURSE:csa1511_cloud computing

Assessment Tasks (Short Answer Test)

1. Define Cloud Computing and Explain Any Four Characteristics That Make It Different from Traditional On-Premise Computing

Definition

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the Internet. Instead of purchasing and maintaining physical hardware, users can access these resources on demand from cloud service providers. This reduces infrastructure costs and provides flexibility to individuals and organizations.

Characteristics

1. On-Demand Self-Service

- Users can access computing resources whenever required without human intervention from the service provider.
- Example: Creating a virtual machine in AWS within minutes.

2. Broad Network Access

- Cloud services are available over the Internet and can be accessed using laptops, smartphones, tablets, or desktops from any location.

3. Resource Pooling

- Cloud providers share computing resources among multiple users using virtualization.
- Resources are allocated dynamically based on user requirements.

4. Rapid Elasticity

- Resources can be increased or decreased automatically according to workload.
- This helps organizations handle traffic spikes efficiently.

Difference from Traditional Computing

- No need to purchase expensive hardware.
- Lower maintenance costs.
- Better scalability and flexibility.

- Accessible from anywhere with an Internet connection.

2. Trace the Evolution of Cloud Computing

Cloud computing has evolved through several technological advancements over the years.

Hardware Evolution

- Computing started with large mainframe computers used by organizations.
- These systems were expensive and required dedicated maintenance.

Personal Computing Era

- Personal computers became affordable, allowing individuals to perform computing tasks independently.

Networking Evolution

- LAN and WAN technologies enabled computers to communicate and share resources.
- Organizations began connecting multiple systems for collaboration.

Internet Software Evolution

- The growth of the Internet enabled web-based applications and online services.
- Technologies like virtualization and distributed computing made it possible to deliver computing resources over the Internet.

Modern Cloud Platforms

Today, platforms like AWS, Microsoft Azure, and Google Cloud provide scalable, secure, and cost-effective cloud services used by millions of businesses worldwide.

3. Explain the Role of Server Virtualization in Cloud Computing

Server Virtualization

Server virtualization is the process of dividing one physical server into multiple virtual servers using a hypervisor. Each virtual machine works independently with its own operating system and applications.

Role in Cloud Computing

- Improves hardware utilization.
- Reduces operational costs.
- Allows multiple users to share physical resources.
- Simplifies backup and disaster recovery.

- Enables quick deployment of virtual machines.

Difference Between Virtualization and Cloud Computing

Virtualization

Technology that creates virtual machines

Runs on physical servers

Improves hardware utilization

Managed within an organization

Cloud Computing

Service model that delivers computing resources

Delivered over the Internet

Provides scalable and on-demand services

Managed by cloud service providers

Examples

- Virtualization: VMware, Hyper-V
- Cloud Computing: AWS, Azure, Google Cloud

4. Discuss the Role of Data Centers in Delivering Cloud Services

A data center is a facility that houses servers, storage devices, networking equipment, power systems, and cooling infrastructure. It forms the backbone of cloud computing by hosting applications and storing user data.

Roles of Data Centers

- Store large amounts of data securely.
- Provide computing resources for cloud applications.
- Ensure continuous availability of services.
- Support backup and disaster recovery.
- Protect data using firewalls, encryption, and physical security.

Advantages

- High reliability
- Better performance
- Secure data management
- Scalable infrastructure
- Supports millions of users simultaneously

5. Differentiate IaaS, PaaS, SaaS, and XaaS with One Suitable Real-World Example

Service Model	Description	Real-World Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage, and networking infrastructure. Users manage the operating system and applications.	Amazon EC2 (AWS)
PaaS (Platform as a Service)	Provides a complete platform for developing, testing, and deploying applications without managing hardware.	Google App Engine
SaaS (Software as a Service)	Delivers software applications over the Internet. Users simply access the software through a browser.	Microsoft 365
XaaS (Anything as a Service)	Refers to delivering any IT service through the cloud, including storage, databases, AI, and security services.	Amazon Web Services (AWS)

Benefits of Cloud Service Models

- Reduces infrastructure costs.
- Improves scalability.
- Faster application deployment.
- Easy software updates.
- Accessible from anywhere.

Conclusion

Cloud computing has revolutionized modern IT by providing flexible, scalable, and cost-effective computing resources over the Internet. Technologies such as server virtualization and data centers enable efficient cloud services, while service models like IaaS, PaaS, SaaS, and XaaS meet the diverse needs of individuals and businesses. Today, cloud computing is widely used in education, healthcare, banking, e-commerce, and many other industries because of its reliability, accessibility, and ease of use.